

Entanglement Gotchas and Equivalence Statements

A Quick-Reference Guide for Two-Qubit Systems

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Purpose

This document collects concise, high-impact facts (“gotchas”) about quantum entanglement. Each statement is phrased as an equivalence, implication, or counterintuitive rule that is commonly used in reasoning, problem-solving, and examinations.

Pure-State Entanglement

- A pure bipartite state is **separable if and only if** it can be written as a tensor product of subsystem states.
- A pure two-qubit state is **separable if and only if** its Schmidt rank is one.
- A pure two-qubit state is **entangled if and only if** it has more than one nonzero Schmidt coefficient.
- A pure two-qubit state is **maximally entangled if and only if** its Schmidt coefficients are equal:

$$\lambda_1 = \lambda_2 = \frac{1}{\sqrt{2}}.$$

- All maximally entangled two-qubit pure states are related by **local unitary transformations**.
- A pure bipartite state is maximally entangled if and only if each subsystem’s reduced density matrix is maximally mixed.
- For pure bipartite states, **all valid entanglement measures induce the same ordering**.

Superposition vs. Entanglement

- A superposition is *not* necessarily entangled.
- Entanglement depends on **correlations between subsystems**, not on the number of terms in a state expansion.
- A state that looks entangled in one basis may appear separable in another; entanglement itself is basis-independent.
- Local basis changes (local unitaries) cannot create or destroy entanglement.

Measurement and Correlation

- A pure bipartite state is entangled if and only if measurement of one subsystem changes the post-measurement state of the other.
- Classical correlation does not imply entanglement.
- Entanglement always implies nonclassical correlation.
- Perfect correlation or anti-correlation in multiple measurement bases is a signature of maximal entanglement.

Mixed States and PPT Criterion

- A mixed quantum state can be correlated without being entangled.
- For 2×2 and 2×3 systems, a state is separable if and only if it has a **positive partial transpose (PPT)**.
- For higher-dimensional systems, PPT is necessary but not sufficient for separability.
- Violation of the PPT condition implies entanglement in all dimensions.
- Some entangled states cannot be distilled into Bell states using LOCC (bound entanglement).

Werner States and Noise

- Werner states are entangled if and only if the mixing parameter satisfies

$$p > \frac{1}{3}.$$

- Werner states may be entangled even when they admit a local hidden variable model.
- Entanglement can disappear abruptly under decoherence (entanglement sudden death).
- Noise typically destroys entanglement before destroying local coherence.

Operational and Resource-Theoretic Gotchas

- Entanglement cannot be increased by **local operations and classical communication (LOCC)**.
- Local operations can transform entanglement but cannot create it from separable states.
- Maximal entanglement does not imply maximal violation of all Bell inequalities.
- Entanglement is necessary but not sufficient for quantum computational advantage.

One-Line Memory Triggers

- “Equal Schmidt coefficients $\Rightarrow$ maximal entanglement.”
- “Product form $\Rightarrow$ no entanglement.”
- “PPT violation $\Rightarrow$ entangled.”
- “Local unitaries do not change entanglement.”
- “Entanglement is about correlations, not superposition.”

Closing Remark

Entanglement is best understood not as a mysterious feature of quantum states, but as a precise statement about how information is distributed across subsystems. The statements collected here serve as a compact map of that structure.